

Assignment 5

September 4, 2026

Given:

- $m_{H^0} \approx 125 \text{ GeV}/c^2$
- $m_{Z^0} \approx 91 \text{ GeV}/c^2$
- $m_e = 5.11 \times 10^{-4} \text{ GeV}/c^2$
- $m_\gamma = 0$

1 Part 1

The maximum energy of the electron at LEP was

$$E = 104 \text{ GeV}.$$

The total relativistic energy of a particle is

$$E = \gamma mc^2.$$

rearranging:

$$\gamma = \frac{E}{mc^2}.$$

rest mass of the electron is

$$m_e = 5.11 \times 10^{-4} \text{ GeV}/c^2.$$

so, solving for γ

$$\gamma = \frac{104 \text{ GeV}}{5.11 \times 10^{-4} \text{ GeV}} \\ \gamma \approx 2.04 \times 10^5.$$

therefore to 1% accuracy,

$$\boxed{\gamma \approx 2.04 \times 10^5}$$

(super close to the speed of light)